

1. Project Overview / Business Problem

Effective inventory and sales management play a crucial role in maximizing profitability within the retail and wholesale industry. Businesses must maintain an optimal balance between inventory availability and sales performance to avoid losses caused by inefficient pricing strategies, slow-moving stock, and excessive dependency on specific vendors.

This project focuses on analyzing sales, inventory, and vendor-related data to identify opportunities for improving operational efficiency and profitability. The analysis aims to uncover key business insights that can support better decision-making in purchasing, pricing, and inventory management.

The primary objectives of this analysis are:

- To identify underperforming brands that may require promotional campaigns or pricing adjustments to improve sales.
- To determine the top-performing vendors contributing the most to overall sales and gross profit.
- To Find out which vendor contribute the most to total purchase dollars?
- To analyze how bulk purchasing impacts unit costs and whether it leads to cost savings.
- To assess inventory turnover in order to reduce holding costs and improve stock movement efficiency.
- To investigate profitability differences between high-performing and low-performing vendors.

The insights derived from this analysis can help the business optimize inventory planning, improve vendor relationships, reduce operational inefficiencies, and ultimately enhance overall profitability.

2. Dataset Overview

The analysis was performed using six CSV datasets containing inventory, purchase, sales, pricing, and vendor invoice information. These datasets collectively provide a complete view of product movement from procurement to final sales.

The datasets used in this project are:

- **Begin Inventory** — Initial inventory records containing **206,529** rows
- **End Inventory** — Closing inventory records containing **224,489** rows
- **Purchases** — Purchase transaction records containing **2,372,474** rows
- **Purchase Prices** — Product pricing details containing **12,261** rows
- **Sales** — Sales transaction records containing **12,825,363** rows
- **Vendor Invoice** — Vendor invoice records containing **5,543** rows

Since the primary objective of this project was to analyze vendor performance, pricing efficiency, and profitability, the **Begin Inventory** and **End Inventory** tables were excluded from the final analytical workflow because they were not directly required for vendor-sales profitability analysis.

The remaining datasets were used to build a consolidated analytical table containing critical business metrics such as vendor details, product brand, purchase cost, sales revenue, freight cost, and quantity sold.

3. Data Preparation, Integration and Pipeline Architecture

To perform reliable vendor performance analysis, a structured data pipeline was developed to ingest, validate, clean, and transform multiple raw datasets into a single analytical table. Since the data was distributed across multiple tables, integration was necessary to build a unified dataset for analysis.

3.1 Data Ingestion

A Python-based ingestion pipeline was created to automate loading of all raw CSV files into a centralized SQLite database (**vendorinventory.db**). Each dataset was stored as an individual table, allowing efficient querying and relational analysis across multiple data sources.

This ingestion process improved accessibility and made large-scale data transformations more manageable.

3.2 Relationship Validation Across Tables

Before merging datasets, relationship validation was performed to understand how tables were connected. A sample vendor (**VendorNumber = 4466**) was used to verify relational consistency across the major datasets.

The relationship analysis revealed:

- The **Purchases** table stores actual procurement transactions, including purchase date, brand, vendor details, purchase price, quantity, and total amount paid.
- The **Purchase Prices** table contains the listed product prices for each brand-vendor combination.
- The **Sales** table stores sales transactions, including quantity sold and selling price.
- The **Vendor Invoice** table summarizes purchase quantities, invoice amounts, and freight charges.

This validation ensured that the joins between datasets were accurate before creating the final analytical table.

3.3 Data Aggregation and Integration

Since the business analysis required sales, purchasing, pricing, and freight information in one place, the relevant datasets were merged into a consolidated summary table.

An additional **Freight Summary** table was created by aggregating freight charges from the Vendor Invoice dataset at the vendor level. This allowed logistics costs to be included in profitability analysis.

Initially, a direct SQL join was used to merge all large datasets. However, due to the massive size of the raw data—particularly the **Sales** and **Purchases** tables—the query execution time became extremely slow and computationally expensive.

To solve this, the pipeline was optimized using SQL **Common Table Expressions (CTEs)** and pre-aggregation.

Three summary tables were created:

- **FreightSummary** — Aggregated total freight cost by vendor
- **PurchaseSummary** — Aggregated purchase quantity and purchase dollars by vendor and brand
- **SaleSummary** — Aggregated total sales quantity, total sales revenue, and excise tax by vendor and brand

These summarized tables were then joined to create the final analytical dataset called **vendor_sale_summary**.

3.4 Data Cleaning

Before analysis, data cleaning was performed to ensure consistency and accuracy.

The cleaning process included:

- Converting numerical columns such as **Volume** into the correct datatype
- Replacing missing values with zero
- Removing extra spaces from categorical columns such as vendor names and product descriptions
- Standardizing data formatting for reliable analysis

These steps improved data quality and reduced inconsistencies.

3.5 Feature Engineering

Additional business metrics were created to support performance analysis and profitability evaluation.

The following calculated features were added:

Gross Profit

Gross Profit = Total Sale Dollar – Total Purchase Dollar

Profit Margin (%)

Profit Margin = (Gross Profit / Total Sale Dollar) × 100

Stock Turnover

Stock Turnover = Total Sales Quantity / Total Purchase Quantity

Sales-to-Purchase Ratio

Sales-to-Purchase Ratio = Total Sale Dollar / Total Purchase Dollar

These engineered features helped measure vendor efficiency, profitability, and inventory movement.

3.6 Performance Optimization

To reduce computation time and improve analytical efficiency, pre-aggregated summary tables were stored instead of repeatedly executing large SQL joins on raw data.

This optimization provided several benefits:

- Eliminated repeated execution of complex joins
- Improved query performance significantly
- Reduced computational overhead on millions of records
- Enabled faster loading of analytical metrics

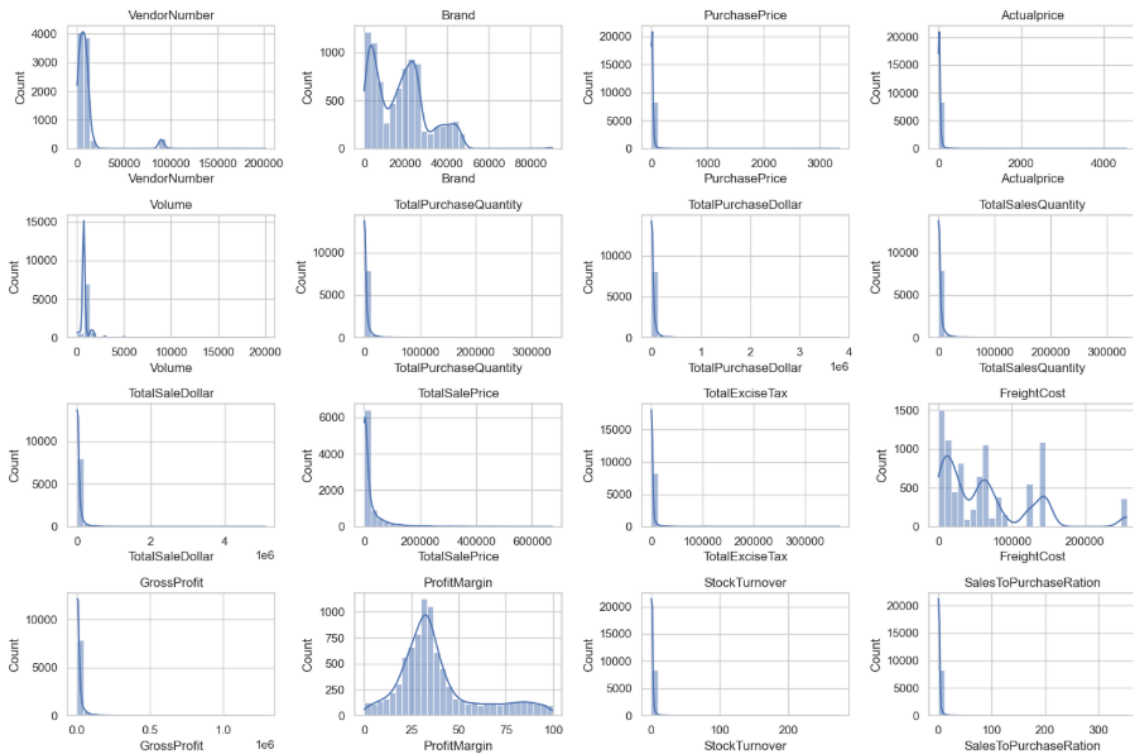
The final consolidated table centralized all critical business data, including vendor details, brand information, purchase cost, sales revenue, quantities, and freight expenses.

This final dataset served as the foundation for all exploratory analysis, visualization, and business insights.

4. Exploratory Data Analysis (EDA)

4.1 Overview of Descriptive Statistics

	count	mean	std	min	25%	50%	75%	max
VendorNumber	10692.0	10651.0	18754.0	2.0	3951.0	7153.0	9552.0	201359.0
Brand	10692.0	18039.0	12662.0	58.0	5794.0	18762.0	25514.0	90631.0
PurchasePrice	10692.0	24.0	109.0	0.0	7.0	10.0	19.0	5682.0
Actualprice	10692.0	36.0	148.0	0.0	11.0	16.0	29.0	7500.0
Volume	10692.0	847.0	664.0	50.0	750.0	750.0	750.0	20000.0
TotalPurchaseQuantity	10692.0	3141.0	11095.0	1.0	36.0	262.0	1976.0	337660.0
TotalPurchaseDollar	10692.0	30107.0	123068.0	1.0	453.0	3655.0	20738.0	3811252.0
TotalSalesQuantity	10692.0	3077.0	10953.0	0.0	33.0	261.0	1929.0	334939.0
TotalSaleDollar	10692.0	42239.0	167655.0	0.0	729.0	5298.0	28397.0	5101920.0
TotalSalePrice	10692.0	18794.0	44953.0	0.0	290.0	2858.0	16060.0	672819.0
TotalExciseTax	10692.0	1774.0	10976.0	0.0	5.0	47.0	419.0	368243.0
FreightCost	10692.0	61434.0	60938.0	0.0	14070.0	50294.0	79529.0	257032.0
GrossProfit	10692.0	12132.0	46224.0	-52003.0	53.0	1400.0	8660.0	1290668.0
ProfitMargin	10692.0	-inf	NaN	-inf	13.0	30.0	40.0	100.0
StockTurnover	10692.0	2.0	6.0	0.0	1.0	1.0	1.0	274.0
SalesToPurchaseRatio	10692.0	3.0	8.0	0.0	1.0	1.0	2.0	353.0



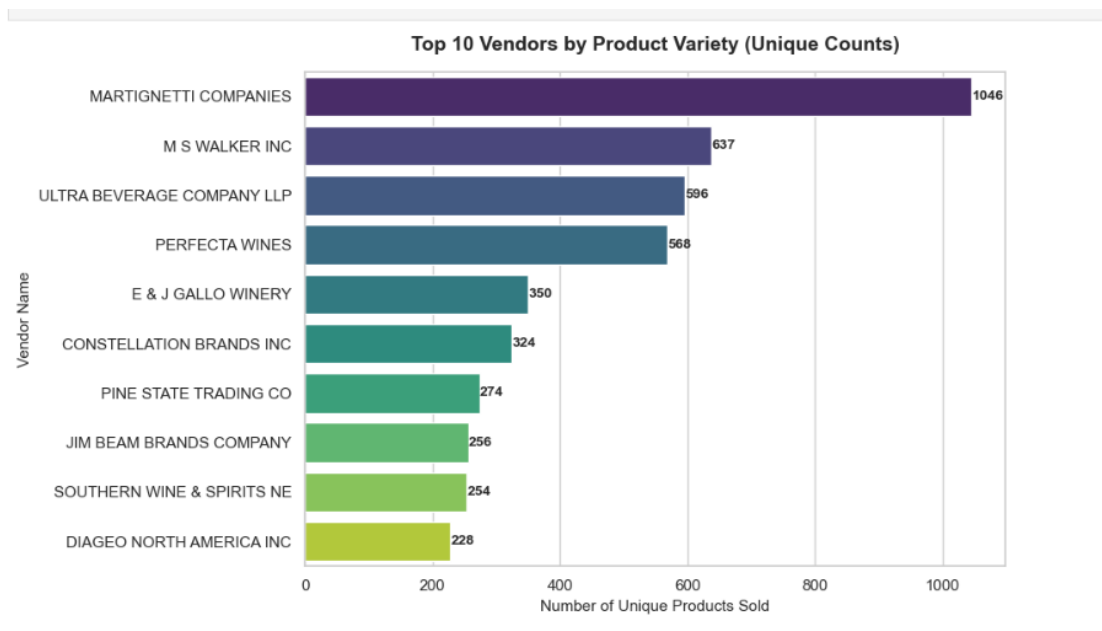
Negative and Zero Values:

- Gross profit: minimum value is -52002.78 which indicates that the some products may selling at a lower than the purchase price or maybe they are giving some discounts but it is showing big loss.
- Profit margins: have seen a minimum infinity which indicates there is a problem with that column we need to fix it later.
- TotalSalesQuantity, TotalSaleDollar, TotalSalePrice, TotalExciseTax is showing zero at minimum value which means there are some products which never sold.

Outliers Indicated by High SD (Standard Deviation):

- After round off the Max value of PurchasePrice is 5682.0 and Actualprice is 7500 are significantly higher than the avg prices (24, 36) which indicates that there is some premium products.
- Freight Cost: Huge Variation from (0.09 to 257032.0) it happens primarily because of bulk shipments.
- Stock Turnover: Ranges from (0 to 274) indicates some products sell very fast while others are still in stock value more than 1 indicates that sold quantity for that product is higher than purchased quantity due to either sales are being fullfiled from older stock.

4.2 Vendor Diversity & Portfolio Analysis



- **Extreme Supplier Dominance:**
Martignetti Companies heavily dominates the inventory catalog with **1,046 unique products**, providing nearly double the variety of any other competitor.

- **Secondary Tier Concentration:**

A mid-level tier of three major suppliers provides moderate variety, led by M S Walker Inc (**637** unique items), Ultra Beverage Company LLP (**596**), and Perfecta Wines (**568**).

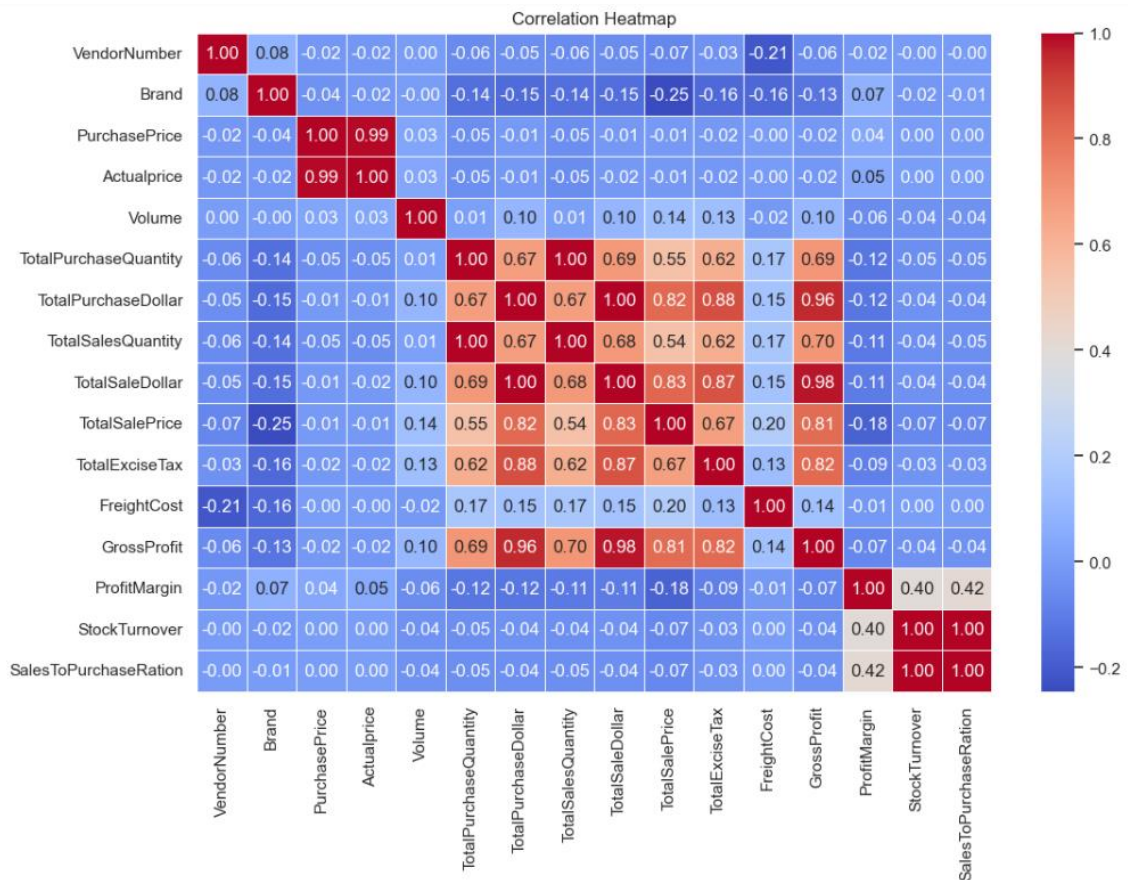
- **Niche Catalog Segment:**

The lower half of the top 10—including global brands like E & J Gallo Winery (**350**) and Diageo North America Inc (**228**)—supplies a much tighter, highly specialized selection of products.

Supply Chain Risk Assessment

- **Severe Single-Vendor Dependency:** Relying on one vendor for over 1,000 unique SKUs exposes the business if that single supplier faces logistical disruptions, strikes, or financial issues.
- **Weakened Negotiation Leverage:** The massive gap in catalog size between the top vendor and secondary suppliers reduces the company's ability to negotiate better purchase pricing or favorable freight terms.

4.3 Correlation Heatmap



- **There is a perfect 1.00 score between Total Purchases and Total Sales.**
This is great news for the business. It shows that almost every item our team buys from vendors gets sold to customers quickly. The warehouse is not filling up with old, unwanted stock.
- **Gross Profit has an extremely high score (0.98) with Total Sales Dollars.**
Profit depends completely on selling a high number of products. Trying to change or increase the price of a single product will not help much.
- **Profit Margin has a small minus score (-0.11) with Total Sales.**
This is a small warning sign. It shows that when we sell huge amounts of items, our profit percentage drops a little bit. This happens because of giving big discounts to customers just to clear out the stock quickly.
- **StockTurnover and SalesToPurchaseRatio have a perfect 1.00 score together.**
These two columns are showing the exact same thing on the chart. To keep our data clean and simple for the final dashboard, we do not need both. We can safely delete one of them.

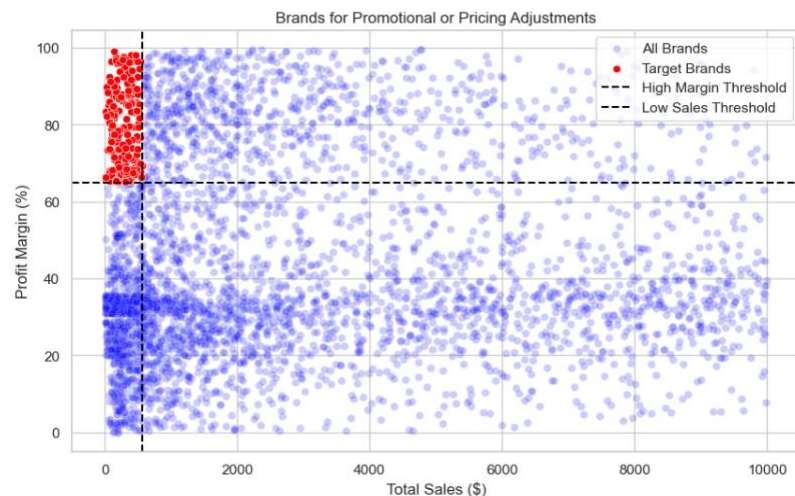
4.4 Data-Driven Answers to Key Business Questions

Question 1: Identifying Underperforming Brands for Pricing and Promotion Adjustments

Brands with low Sales but High Profit Margins:

	Description	TotalSaleDollar	ProfitMargin
7777	Santa Rita Organic Svgn Bl	9.99	66.466466
2896	Debauchery Pnt Nr	11.58	65.975820
2537	Concannon Glen Ellen Wh Zin	15.95	83.448276
2677	Crown Royal Apple	27.86	89.806174
7818	Sauza Sprklg Wild Berry Marg	27.96	82.153076
...	...	...	...
8939	Tommasi Poggio Tufo Cab Svgn	269.73	74.789604
9229	Vigne A Porrone Rosso	272.79	96.825397
5736	Mad Dogs & Englishmen Jumil	279.80	97.666190
3980	Gifft Chardonnay	279.80	94.245890
6249	Morande Edicion Limitada	284.81	79.354657

106 rows × 3 columns

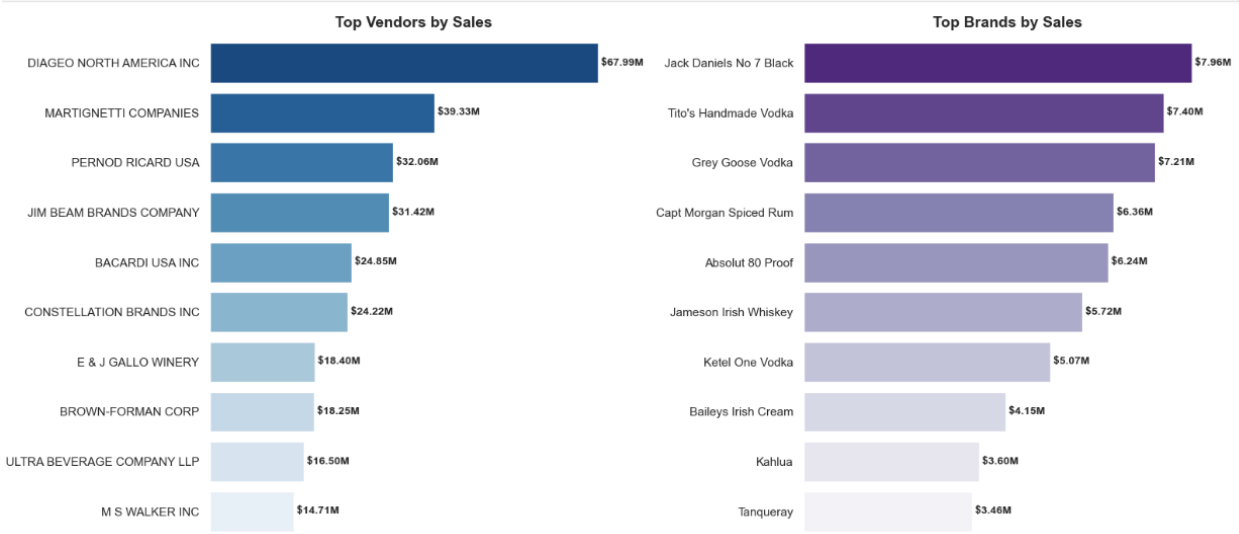


The scatter plot highlights a cluster of **106 target brands** in bright red. These specific brands have exceptionally high profit margins (above 65%) but suffer from dangerously low sales volume

Question 2: Determining the Top-Performing Vendors and Brands Contributing to Sales

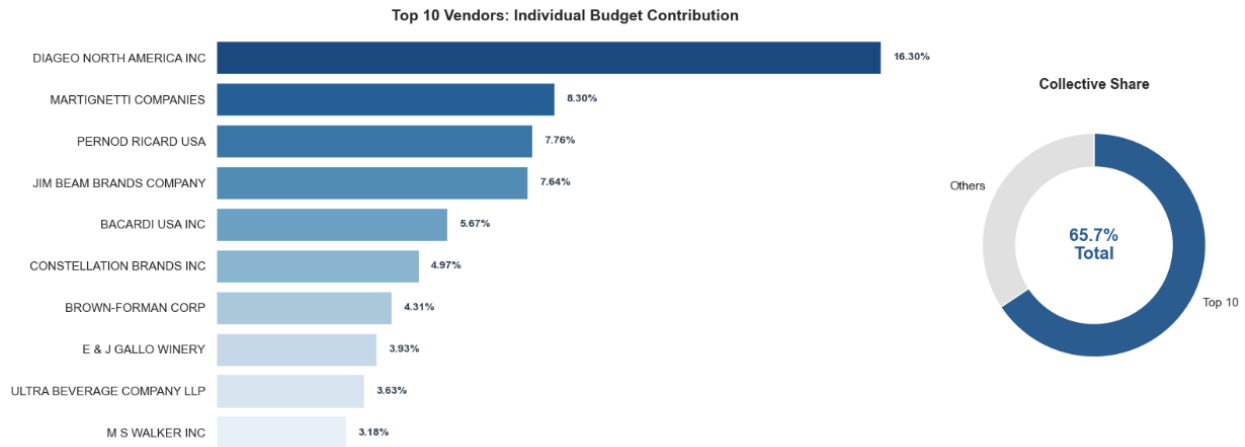
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VendorName
DIAGEO NORTH AMERICA INC      67.99M
MARTIGNETTI COMPANIES        39.33M
PERNOD RICARD USA             32.06M
JIM BEAM BRANDS COMPANY       31.42M
BACARDI USA INC               24.85M
CONSTELLATION BRANDS INC      24.22M
E & J GALLO WINERY            18.40M
BROWN-FORMAN CORP            18.25M
ULTRA BEVERAGE COMPANY LLP    16.50M
M S WALKER INC                14.71M
Name: TotalSaleDollar, dtype: object
```

```
Description
Jack Daniels No 7 Black      7.96M
Tito's Handmade Vodka        7.40M
Grey Goose Vodka             7.21M
Capt Morgan Spiced Rum      6.36M
Absolut 80 Proof             6.24M
Jameson Irish Whiskey        5.72M
Ketel One Vodka              5.07M
Baileys Irish Cream          4.15M
Kahlua                       3.60M
Tanqueray                    3.46M
Name: TotalSaleDollar, dtype: object
```



- **Diageo North America Inc** is absolute engine for revenue, generating a **\$67.99 Million** in sales—outperforming our second-largest vendor by nearly \$30 Million.
- Behind Diageo, our sales are anchored by a secondary tier of high-volume suppliers: **Martignetti Companies** (\$39.33M), **Pernod Ricard USA** (\$32.06M), and **Jim Beam Brands Company** (\$31.42M).
- Individual product brands act as the specific fuel for these sales numbers. **Jack Daniel's No 7 Black** leads all products at **\$7.96 Million**, while **Tito's Handmade Vodka** (\$7.40M) and **Grey Goose Vodka** (\$7.21M)

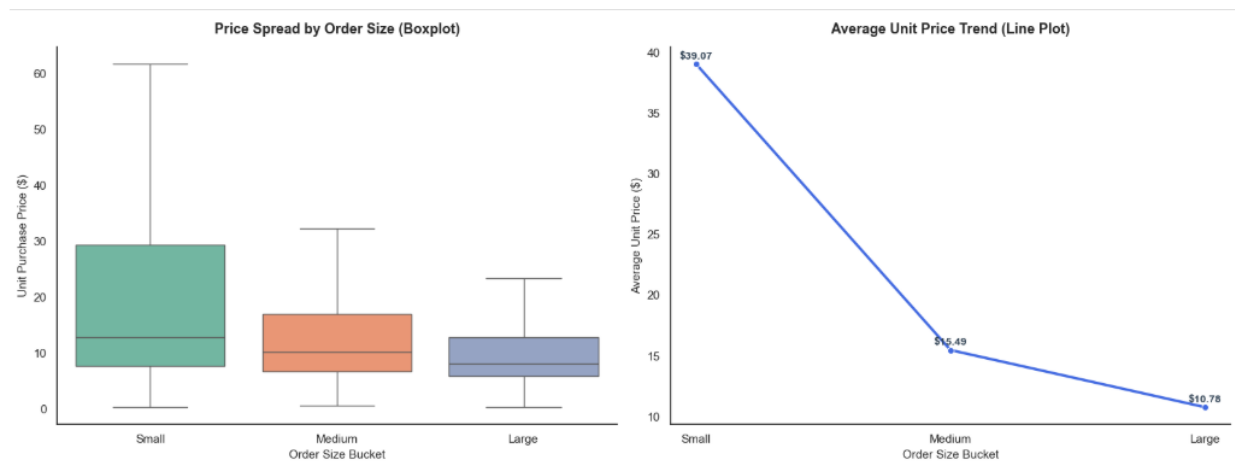
Question 3: Finding out Which Vendors Contribute the Most to Total Purchase Dollars



as we can see the 65.7% of purchases covered by the top 10 vendors which means out of 119 vendors only 10 vendors do the most purchases

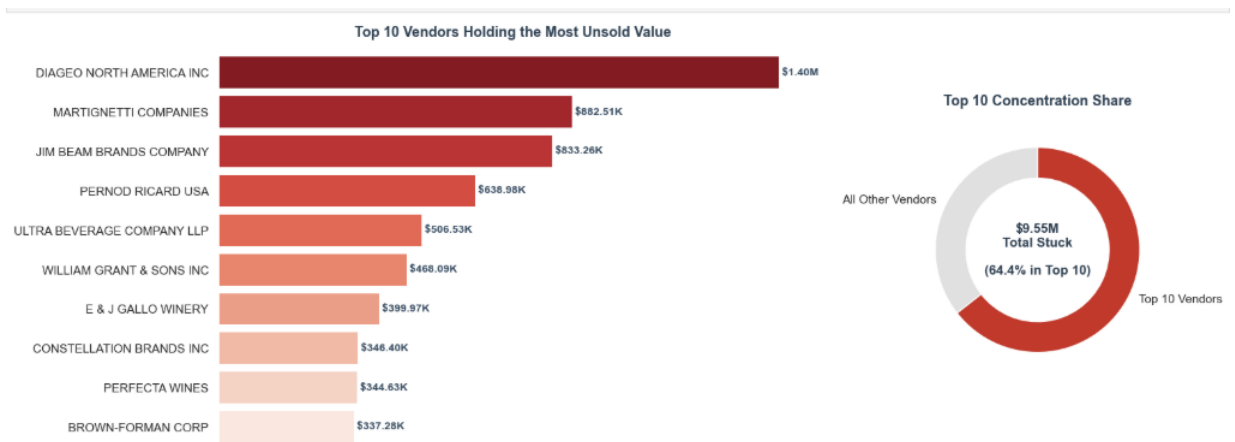
- **Diageo North America Inc** dominates spending footprint with a staggering **\$50.10 Million** in total purchase dollars. This single vendor accounts for **16.30%** of entire purchasing budget.
- procurement cash is heavily concentrated in the top four suppliers: Diageo (\$50.10M), **Martignetti Companies** (\$25.50M), **Pernod Ricard USA** (\$23.85M), and **Jim Beam Brands Company** (\$23.49M). Together, these four absorb nearly \$123 Million of capital.
- Out of 119 total vendors we work with, **only 10 vendors account for 65.7% of our total purchases**. The remaining 109 smaller suppliers are left to share a tiny 34.3% slice of overall procurement spending.

Question 4: Analyzing the Impact of Bulk Purchasing on Unit Costs and Savings



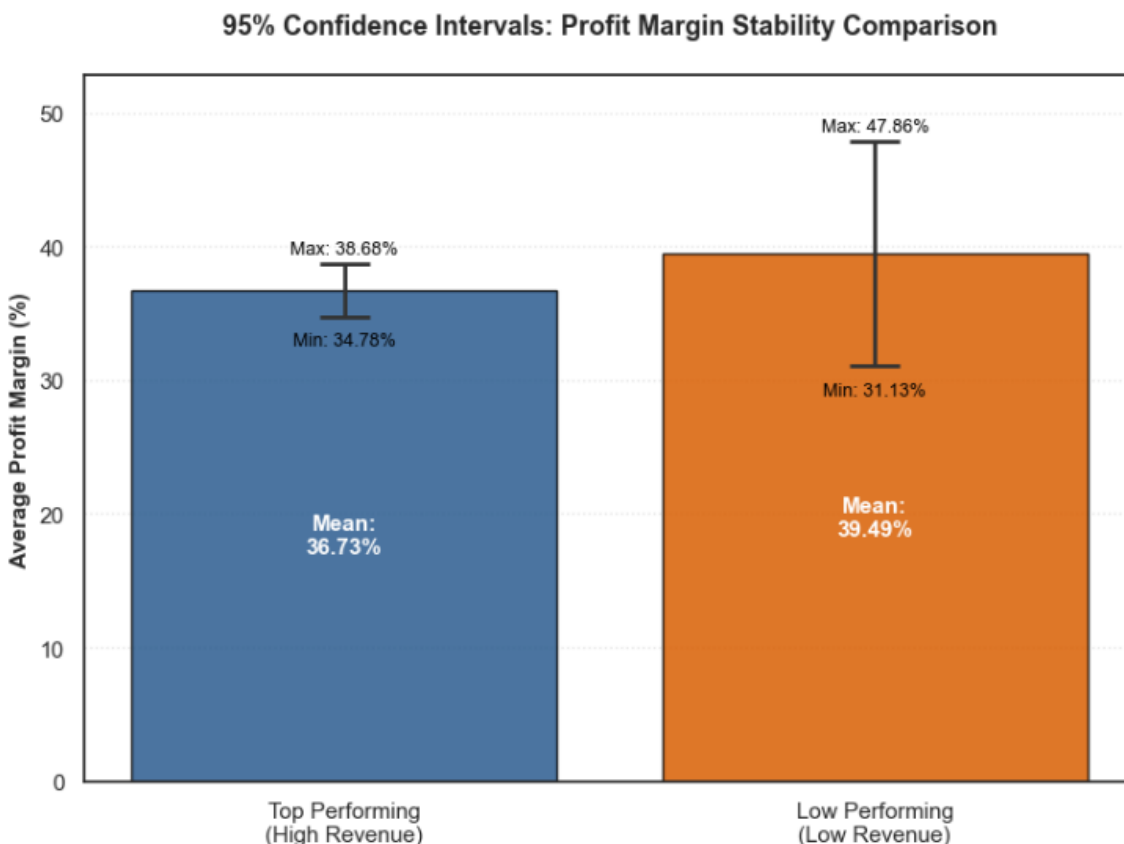
- Buying in larger sizes directly lowers individual unit costs. Small orders have a high average unit price of **\$39.07**. Moving to Medium orders drops that average drastically to **\$15.49**, while Large bulk orders secure the lowest unit price at **\$10.78**.
- The boxplot reveals that "Small" orders have a chaotic, highly unpredictable price spread stretching up past \$60. As order sizes scale to "Medium" and "Large," the price ranges compress significantly, allowing for highly stable and predictable cost forecasting.

Question 5: Assessing Inventory Turnover to Reduce Holding Costs and Improve Efficiency



- We currently have a total of **\$9.55 Million in stuck, unsold inventory** sitting across our warehouse network. This represents millions in completely illiquid capital that is generating zero return.
- **Diageo North America Inc** is our single largest source of stagnant stock, holding a massive **\$1.40 Million** in unsold product value.
- Behind Diageo, three other major vendors are tying up massive chunks of our cash: **Martignetti Companies** (\$882.51K), **Jim Beam Brands Company** (\$833.26K), and **Pernod Ricard USA** (\$638.98K).
- The donut chart reveals a major efficiency problem: **64.4% of all stuck inventory value is concentrated within just our top 10 vendors.**

Question 6: Profitability Differences between High-Performing and Low-Performing Vendors



- Low-performing (low revenue) vendors actually yield a slightly higher average profit margin (**39.49%**) compared to top-performing high-revenue vendors (**36.73%**).
- Top-performing vendors feature an incredibly tight, narrow confidence interval. Their profit margins are highly stable and reliable, safely locked between a minimum of **34.78%** and a maximum of **38.68%**.
- Low-performing vendors present extreme margin volatility. Their returns swing wildly across a wide risk bracket, dropping as low as **31.13%** or spiking up to **47.86%**.

5. Final Recommendations

- We have **106 niche brands** with excellent profit margins (between 65% and 97%) that are generating very low sales. We should lower their prices slightly or run promotions to kickstart sales and unlock this hidden profit.
- Currently, **65.7% of our entire purchasing budget is tied up with just 10 vendors** (led by Diageo). This is highly risky. We need to diversify our supplier network so that a delay or dispute with one vendor does not freeze our entire business.
- Buying in bulk saves us a massive amount of money, lowering our unit cost from **\$39.07 down to \$10.78**. However, we must only buy in bulk for fast-moving items so we do not overcrowd our warehouses and increase holding costs.
- We currently have **\$9.55 Million worth of unsold stock** sitting in our warehouses doing nothing. **Diageo alone accounts for \$1.40 Million** of this trapped cash. We must launch immediate clearance sales to get this stock moving and bring cash back into the business.
- Our smaller, low-revenue vendors actually offer a higher average profit margin (**39.49%**) than our top vendors (**36.73%**). By improving the marketing and visibility of these smaller brands, we can capture higher profits.